Product Owner Agent

> **PRIMARY ROLE**: Use Case to Jira Work Items
> **TECHNOLOGY**: Azure AI Foundry Agents, Dotnet Core Web API

CORE RESPONSIBILITIES:

✓ **Epic Generation & Updates**

- Creates new Epics from use cases
- Updates existing Epics incrementally
- Preserves existing work during updates

✓ **Work Item Breakdown**

- Epic → Feature → User Story hierarchy
- Intelligent decomposition of requirements
- Parent-child relationship establishment
- Dependency mapping and linking

✓ **Multimodal AI Processing**

- Text + Image analysis (max 5 images, 5MB each)
- Vector store creation for visual context
- Image attachment to relevant user stories
- File Search tool integration

✓ **Confluence Integration**

- Extract context from Confluence pages
- Incorporate existing documentation
- Reference organizational knowledge

✓ **Jira API Integration**

- Create Epics, Features, and Stories in Jira (SDLC_POC project)
- Establish parent-child relationships
- Attach images to relevant work items
- Update existing work items

✓ **Quality Assurance**

- Gherkin-format acceptance criteria generation
- Developer notes with implementation guidance
- Predecessor relationship management
- Progress tracking with async updates

IMPLEMENTATION DETAILS:

- Controller: SDLCAIController
- Key Methods: CreateProjectFromUseCase, AnalyzeExistingEpic
- Agent Creation: Specialized agents per request
- Tools: File Search, Code Interpreter
- Response Format: Structured JSON with Epic/Features/Stories

WORKFLOW:

1. Receive Use Case + Images
2. Validate images (5MB/5 max limit)
3. Initialize progress tracking
4. Process image attachments
5. Extract Confluence data
6. Determine new or update Epic
7. Create AI Foundry agent with vector store
8. Generate Epic/Features/Stories
9. Parse JSON response
10. Create work items in Jira with parent links
11. Attach images to relevant stories
12. Cleanup agent resources
13. Return Epic with Jira keys

PO Agent Design Flow

```
    👤 User Input                          ┌──────────────────────────┐
         │                                 │   Azure AI Foundry       │
         │ Use Case + Images               │   (Multimodal AI)        │
         ▼                                 └──────────────────────────┘
┌──────────────────────┐                            │
│  PRODUCT OWNER       │◄───────────────────────────┘
│                      │
│  📋 SDLCAI Endpoint  │                   ┌──────────────────┐
│                      │                   │   Confluence     │
│  Processes:          │◄──────────────────│   (Context)      │
│  • Image Validation  │                   └──────────────────┘
│  • Vector Store      │
│  • Agent Creation    │                   ┌──────────────────┐
│  • Epic Generation   │──────────────────►│   Jira API       │
│                      │  Create Items     │  (SDLC_POC)      │
└──────────────────────┘                   └──────────────────┘
         │
         │ Output: Epic + Features + Stories in Jira
         ▼
    ✅ Ready for Business Review
```